MicroRNA Regulation in Embryonic Transitions

Abstract

Deciphering microRNAs' (miRNAs) maternal-to-zygotic transition regulatory roles in *Drosophila melanogaster* is still one of the developing biology's critical issues. This study aimed to identify differentially expressed miRNAs between 2-hour and 5-hour *Drosophila melanogaster* embryos and investigate their regulatory roles in this developmental transition. Differential expression analysis was done using RNASeq data with DESeq2, followed by target prediction with TargetScanFly and functional and pathway enrichment using FlyEnrichr. The results revealed that the up-regulated miRNA, dme-miR-3-3p, is enriched in pathways of transcriptional activation and cytoskeletal reorganization, events critical for the onset of zygotic genome activation. On the other hand, several down-regulated miRNAs like dme-miR-315-5p were associated with maternal mRNA stabilization and cellular signaling, highlighting their roles in reducing maternal contribution while promoting autonomous development. Functional and pathway enrichment analysis confirmed regulatory roles for these miRNAs in transcriptional and structural pathways. The results strongly support that miRNAs are the primary regulators of developmental shifts and that they are the key players in the initial stages of embryogenesis to which this study aims to contribute to understanding the role of miRNAs during this essential phase.

Introduction

MicroRNAs (miRNAs) are small, non-coding RNA molecules that regulate gene expression by targeting messenger RNAs (mRNAs) at the post-transcriptional level[1]. This regulation occurs through mechanisms such as mRNA degradation or inhibition of protein translation[2]. These regulatory entities are fundamental in controlling critical cellular functions, including differentiation, development, and maintaining cellular homeostasis[3]. In the model organism *Drosophila melanogaster*, miRNAs play an especially pivotal role during early embryogenesis[4]. Specifically, they facilitate the transition from maternally derived transcripts to zygotically driven transcription, a step essential for initiating key developmental programs[5].

The precise regulation of miRNAs is critical for coordinating developmental processes[6]. Up-regulated miRNAs are often involved in activating essential pathways, including transcriptional initiation and cytoskeletal reorganization[7,8]. These activities are necessary to establish new cellular programs during development[9]. In contrast, down-regulated miRNAs serve to attenuate maternal regulatory influences, thereby reducing dependence on earlier developmental frameworks. This down-regulation enables the embryo to achieve autonomous regulatory control[10,11,12]. Together, the dynamic interplay between up and down-regulated miRNAs ensures the successful progression of developmental transitions, such as the activation of the zygotic genome[5,13].

While understanding of miRNA mechanisms has progressed considerably, roles of such small RNAs in developmental transitions are as yet incompletely defined[14]. As an example, the functional effects of differentially regulated miRNAs across the maternal-to-zygotic transition in *Drosophila melanogaster* are yet to be characterized[9]. Embryogenesis has two major time points of interest; 2–3 hr post-fertilization is when

maternal transcripts dominate and at ~ 5 hr where zygotic transcription begins to take place. These stages offer a unique opportunity to investigate miRNA-mediated regulation of processes during early development[15].

This investigation aims to identify differentially expressed miRNAs between these two stages of development and elucidate their biological function and regulatory pathways. The up-regulated miRNAs at 5 hours post-fertilization are suggested to transcribe activators of zygotic genome activation and cytoskeletal pathways, and the down-regulated miRNAs are predicted to alleviate maternal regulatory effects to activate the development autonomously.

Methods

Data Collection

Drosophila melanogaster RNASeq data were obtained from the Sequence Read Archive (SRA) database[16], focusing on two development stages: stage 1 (2-3 hr embryos) and stage 2 (~5 hr embryos). Each stage included two replicates (SRA accessions: SRR5935244, SRR5935245 for 2 hr; SRR5935247, SRR5935249 for 5 hr). Data was downloaded and processed with the Galaxy platform (https://usegalaxy.org). Reads from SRA files were extracted in FASTQ format using Galaxy's "Download and Extract Reads in FASTQ format from NCBI SRA" tool[17].

Read Preprocessing and Mapping

Raw sequencing data underwent quality trimming to eliminate adapter sequences using the "Clip" tool in Galaxy, with "TGGAATTCTC" specified as the adapter sequence. The processed reads were then aligned to the Drosophila melanogaster genome assembly (BDGP Release 6 + ISO MT/dm6) through Bowtie2 with its default configurations[18]. BAM files were subsequently created from the aligned reads for downstream analysis and visualization.

Differential Expression Analysis

The htseq-count tool in Galaxy quantified miRNA expression using genome annotations from miRBase (dme.gff3)[19]. Differential expression analysis was performed with DESeq2 in Galaxy, and the output files were downloaded for further analysis[20]. Processed data in R (https://www.R-project.org/) identified miRNAs with significant differential expression, determined by adjusted p-values $\leq$ 0.05. Expression changes were characterized using log2 fold change (log2(FC)), with positive and negative values representing up and down-regulation respectively. A volcano plot generated in R to visualize miRNA distribution based on log2(FC) and adjusted p-values. These filtered miRNAs were then prepared in R for target prediction[21].

Target Prediction

TargetScanFly (https://www.targetscan.org/fly_72/) was used to predict transcripts targeted by differentially expressed miRNAs[22]. MiRNAs from the differential expression analysis were entered into the tool to generate output tables of predicted targets, then filtered in R and retained high-confidence targets with an Aggregate PCT score ≥ 0.5 or a Cumulative Weighted Context Score ≤ -0.3. Filtered target transcripts lists were extracted and saved for functional and pathway enrichment analyses.

Functional and Pathway Enrichment Analysis

FlyEnrichr (https://maayanlab.cloud/FlyEnrichr/) was used for functional and pathway enrichment analyses by inputting files generated from target prediction[23]. Functional enrichment utilized the GO Molecular Function 2018 database, ranking terms by Combined Score and limiting ontology levels to four for specificity. Pathway enrichment was conducted using the InterPro Domains 2019 database, similarly ranked by Combined Score. Both outputs were visualized as bar plots in R, showing the top five terms for each miRNA.

Results

Differential Expression of miRNAs

Differential expression analysis revealed significant changes in miRNA expression between 2-hour and 5-hour Drosophila embryos. The up-regulated dme-miR-3-3p exhibited a positive log2 fold change (Figure 1), indicating its increased expression at 5 hours, which suggests its involvement in activating zygotic transcription and early developmental pathways[9,15]. Meanwhile, 11 miRNAs, especially dme-miR-315-5p, were significantly down-regulated with negative log2 fold changes (Figure 1), indicating a reduction in maternal regulatory influence as the embryo transitions to zygotic control[24].

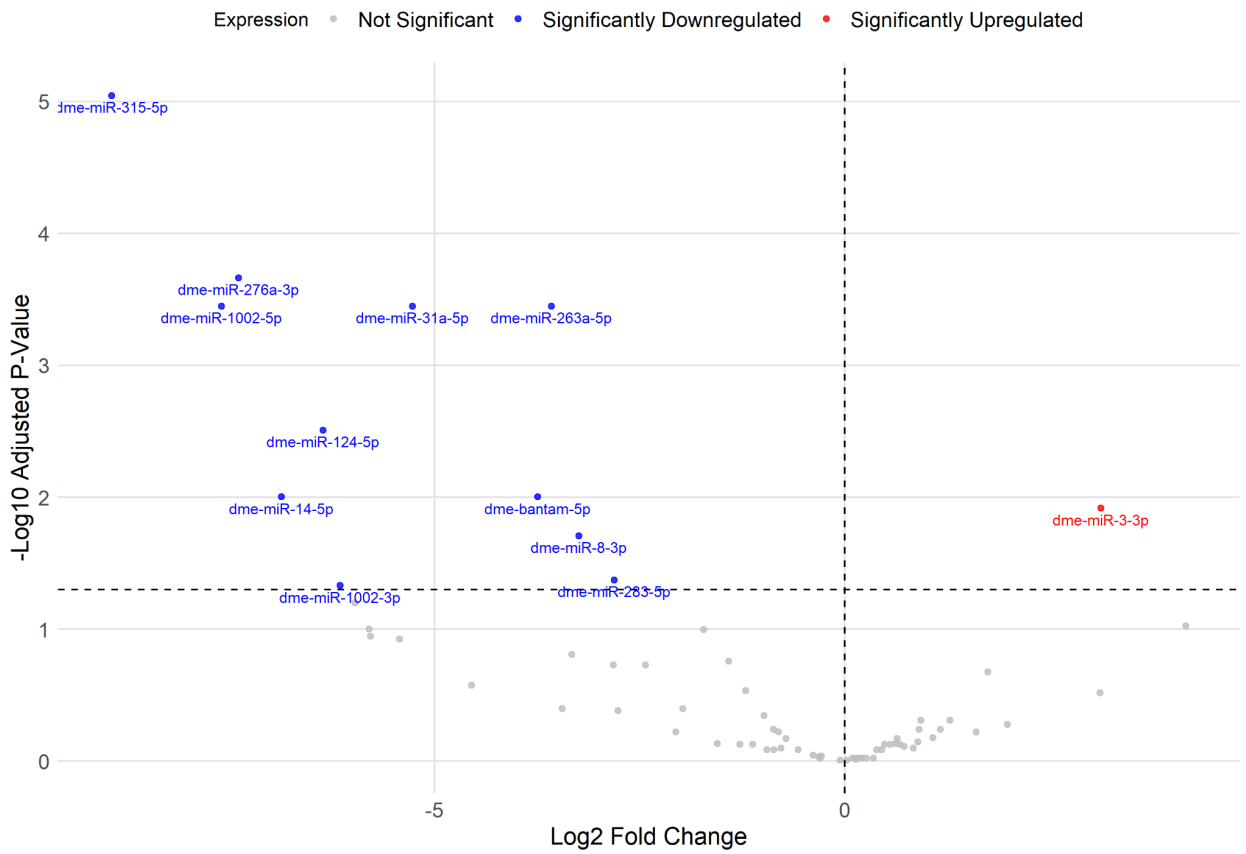

Figure 1: Volcano plot of differentially expressed miRNAs. Up-regulated miRNAs are highlighted in red, down-regulated miRNAs in blue, and non-significant miRNAs in gray. Dashed lines represent significance thresholds for adjusted p-values and log2 fold changes.

Functional and Pathway Enrichment of miRNA Targets

Enrichment analyses revealed distinct roles for differentially expressed miRNAs in regulating developmental processes. The up-regulated dme-miR-3-3p was linked to transcriptional activation and pathways involving SH3 domains(Figures 2 and 3), which mediate protein-protein interactions and cytoskeletal organization essential for early development, suggesting its role in initiating zygotic genome activation and structural coordination[9,15,25]. In contrast, down-regulated miRNAs were associated with maternal regulatory processes that diminish in importance during the transition to zygotic control(Figures 2 and 3). For instance, dme-miR-315-5p was enriched for mRNA binding and ion transport, reflecting its involvement in stabilizing maternal transcripts and cellular signaling[9,24]. Other down-regulated miRNAs, such as dme-bantam-5p, dme-miR-1002-3p, and dme-miR-276a-3p, were linked to RNA processing, ion channel activity, and GPCR signaling, respectively. Additionally, miRNAs like dme-miR-31a-5p and dme-miR-263a-5p were enriched for processes such as apoptosis and extracellular matrix remodeling, highlighting their roles in cell fate determination and structural reorganization[26](Figures 2 and 3).

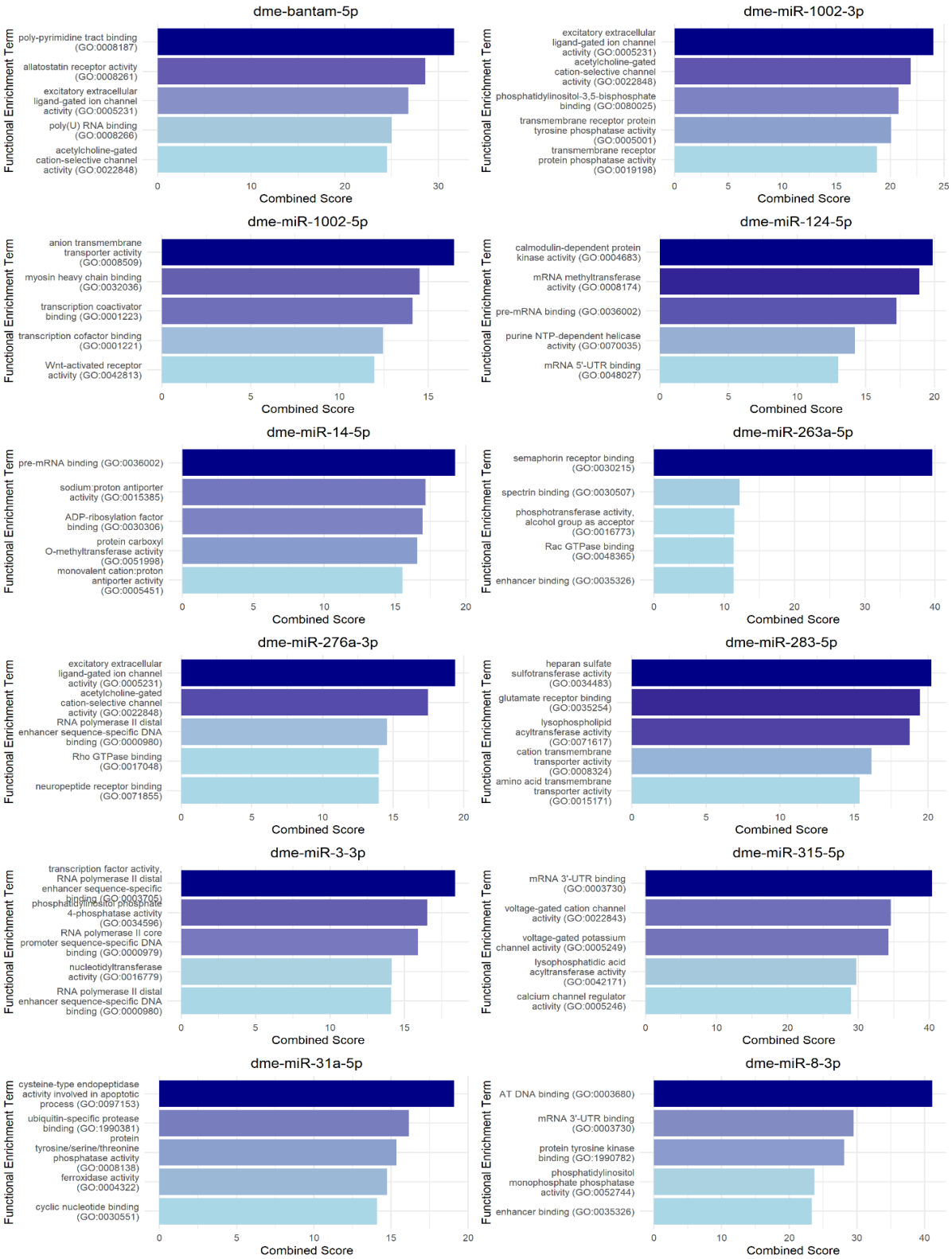

Figure 2: Functional enrichment analysis for up and down-regulated miRNAs. Bar plots show the top enriched biological functions associated with each miRNA, ranked by combined enrichment scores. The gradient represents the intensity of the combined score, with darker bars indicating higher combined score.

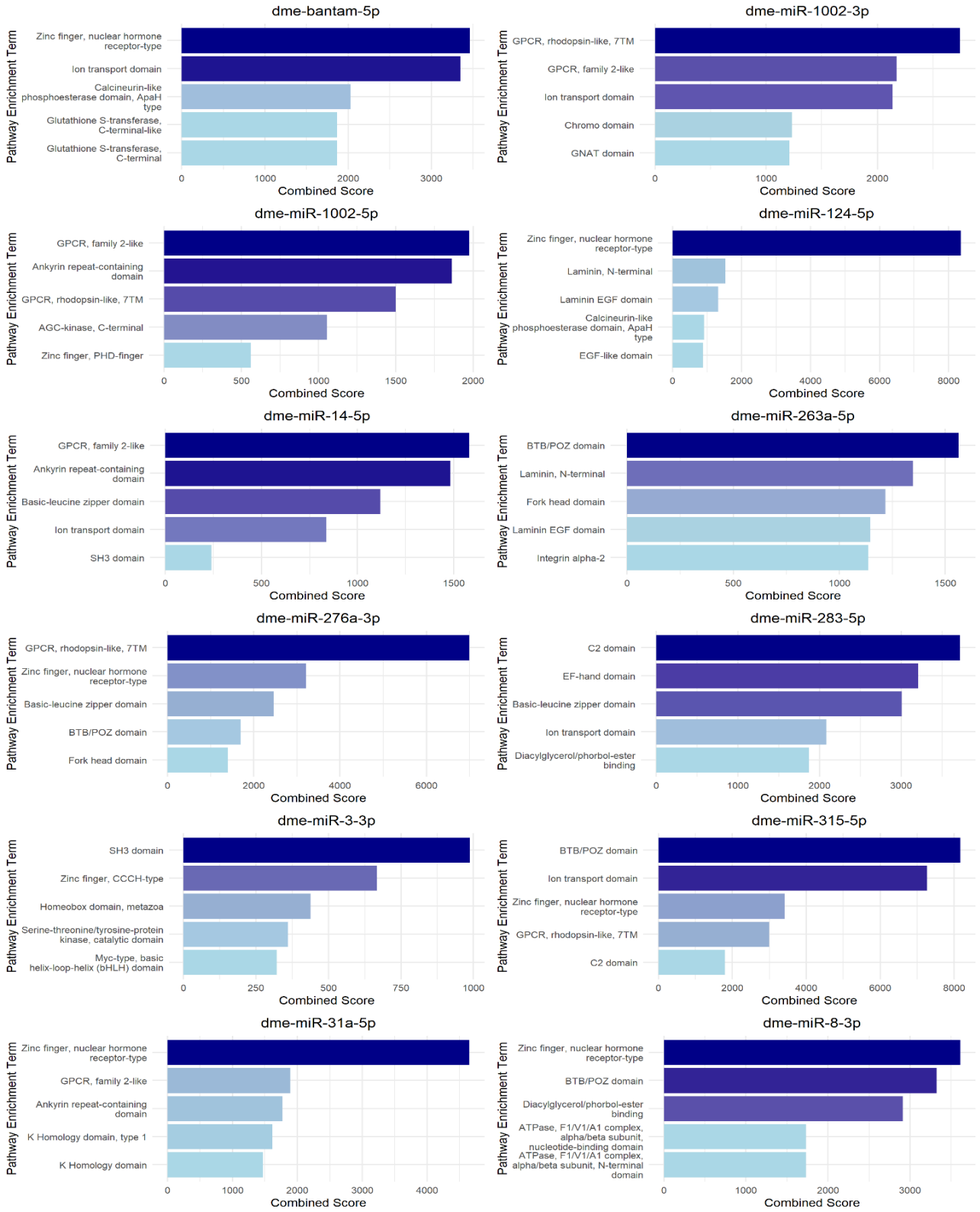

Figure 3: Pathway enrichment analysis for up- and down-regulated miRNAs. Bar plots display the top enriched pathways for each miRNA, ranked by combined enrichment scores. The gradient represents the intensity of the combined score, with darker bars indicating higher combined score.

Discussion

This study investigated the roles of differentially expressed miRNAs during the transition from maternal to zygotic control in *Drosophila melanogaster* embryogenesis. The findings support the hypothesis that up-regulated miRNAs at 5 hours post-fertilization activate zygotic transcription and cytoskeletal organization, while down-regulated miRNAs alleviate maternal influences to enable autonomous development. The up-regulated dme-miR-3-3p was enriched for pathways related to transcriptional activation and cytoskeletal reorganization, consistent with its role in initiating zygotic genome activation[9,15,25]. This agrees with prior research emphasizing miRNA roles in transcriptional regulation and structural coordination during developmental transitions[7,9]. Down-regulated miRNAs were associated with processes like mRNA binding, ion transport, and apoptosis, consistent with their roles in reducing maternal effects and stabilizing cellular signaling[10,11,26].

This study's reliance on computational filtering criteria in R introduces potential biases. High-confidence criteria (Aggregate PCT $\geq$ 0.5 or Cumulative Weighted Context Score $\leq$ -0.3) may exclude biologically relevant targets, potentially overlooking key miRNA-regulated pathways. Experimental validation is necessary to complement computational predictions. Similarly, target prediction within silico tools like TargetScanFly may not capture all miRNA-mRNA interactions under specific conditions, and pathway enrichment results depend on the existing database's accuracy. The use of only two time points also simplifies the dynamic nature of miRNA regulation during embryogenesis. Including additional developmental stages in future studies would provide a more comprehensive view of temporal miRNA activity. Investigating interactions between miRNAs and other regulatory molecules, such as transcription factors and long non-coding RNAs, could further elucidate their developmental roles.

Nevertheless, this study highlights the critical functions of miRNAs in early development. By modulating pathways for zygotic genome activation and cytoskeletal reorganization, up-regulated miRNAs ensure proper zygotic transcription initiation. Concurrently, the down-regulation of maternal miRNAs facilitates the transition to autonomous embryo development, aligning with existing literature on the intricate regulatory interplay required for successful developmental transitions[5,28,29].

In conclusion, this research enhances understanding of miRNA-mediated regulation during early embryogenesis in *Drosophila melanogaster*. These findings inform future studies of miRNA functions in other developmental contexts and provide a foundation for investigating miRNA dysregulation in diseases. Experimental validation and further exploration of miRNA interactions will strengthen these findings and uncover additional regulatory mechanisms.

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